Unit: 6 Unit 5: Linear Equations
Topic: Rate of Change and Slope

Name: ____________________
Date: ____________

Foundation (Jalapeno)

Q1. What is the slope of the line $y = 2x + 1$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q2. A basketball player scores y points in x games. If $y = 3x - 2$, what is the rate of change of points scored per game?
(A) 1 point per game
(B) 2 points per game
(C) 3 points per game
(D) 4 points per game
(E) 5 points per game

Q3. The table shows the distance d traveled by a cyclist in t hours. What is the slope of the line?
(A) 10 km/h
(B) 15 km/h
(C) 20 km/h
(D) 25 km/h
(E) 30 km/h

Q4. The line $y = -2x + 5$ represents the score of a tennis player. What is the slope of the line?
(A) -1
(B) -2
(C) 0
(D) 1
(E) 2

Q5. The graph shows the height h of a basketball player after t years. What is the slope of the line?
(A) 0.5 cm/year
(B) 1 cm/year
(C) 1.5 cm/year
(D) 2 cm/year
(E) 2.5 cm/year

Q6. What is the slope of a vertical line?
(A) positive
(B) negative
(C) zero
(D) undefined
(E) 1

Q7. The equation $y = 0.5x + 2$ represents the distance y traveled by a runner in x minutes. What is the rate of change of distance traveled per minute?
(A) 0.25 km/min
(B) 0.5 km/min
(C) 0.75 km/min
(D) 1 km/min
(E) 1.25 km/min

Q8. A football player throws a ball with an initial velocity of v m/s. The equation $d = vt$ represents the distance d traveled by the ball in t seconds. What is the slope of the line?
(A) v
(B) t
(C) d
(D) 1
(E) 0

Open Ended Questions (FRQ)

Q9. The table shows the number of points y scored by a team in x games. Calculate the slope of the line and interpret it as the rate of change of points scored per game. Show your work and explain your answer.

Q10. The equation $y = -3x + 10$ represents the number of yards y a football team has to go to score a touchdown in x plays. Calculate the slope of the line and interpret it as the rate of change of yards to go per play. Show your work and explain your answer.

Chef's Tips

- Tip 1: Ensure students understand the concept of slope and rate of change.
- Tip 2: Emphasize the importance of interpreting slope in the context of real-world problems.
- Tip 3: Encourage students to show their work and explain their answers in the free-response questions.